2025

Time : $1\frac{1}{2}$ hours

Full Marks : 40

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from ***both*** *the sections as directed.*

Section–A

1. Answer any **four** questions of the following : 6×4=24.

   (a) State and prove Archimedean property of real number.

   (b) Define convergent sequence and prove that every convergent sequence is bounded.

(c) Prove that the sequence $\{u_n\}$ converges to a limit lying between 2 and 3 where $u_n = \left(1+\frac{1}{n}\right)^n$.

(d) A Monotonic increasing sequence bounded above tends to a limit which is its least upper bound.

(e) State and Prove Ratio test.

(f) To Prove that the series $a+ar+ar^2+ar^3+.....\infty$ is convergent when $|r|<1$ and divergent when $r \geq 1$.

Section–B

Answer any **four** questions of the following : 4×4=16

2. Show that $\lim_{n\to\infty} n^{\frac{1}{n}} = 1$

3. Prove that the sequence $\sqrt{2}, \sqrt{2+\sqrt{2}}, \sqrt{2+\sqrt{2+\sqrt{2}}}, ...$ converges to 2.

4. Test the convergence of the series whose general term is $u_n = \sqrt{n^4+1} - \sqrt{n^4-1}$.

5. Test the convergence of the series $\sqrt{\frac{1}{2}} + \sqrt{\frac{2}{3}} + \sqrt{\frac{3}{4}} + \sqrt{\frac{4}{5}} + ...$

6. Prove that the series $1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + ...\infty$ is conditionally convergent.

Or

Define absolutely convergent and conditionally convergent with example.

7. Define countable and uncountable sets with two examples.

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